



Lipids and brain inflammation in *APOE4*-associated dementia

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Purpose of review

To highlight recent developments in studying mechanisms by which the apolipoprotein E4 (*APOE4*) allele affects the metabolism of brain lipids and predisposes the brain to inflammation and Alzheimer's disease (AD) dementia.

Recent findings

APOE4 activates Ca^{2+} dependent phospholipase A2 (cPLA2) leading to changes in arachidonic acid (AA), eicosapentaenoic acid and docosahexaenoic acid signaling cascades in the brain. Among these changes, the increased conversion of AA to eicosanoids associates with sustained and unresolved chronic brain inflammation. The effects of *APOE4* on the brain differ by age, disease stage, nutritional status and can be uncovered by brain imaging studies of brain fatty acid uptake. Reducing cPLA2 expression in the dementia brain presents a viable strategy that awaits to be tested.

Summary

Fatty acid brain imaging techniques can clarify how changes to brain polyunsaturated fatty acid metabolism during the various phases of AD and guide the development of small molecules to mitigate brain inflammation.

Keywords

Alzheimer's disease, ApoE4, arachidonic acid, cytosolic phospholipase A2, neuroinflammation

INTRODUCTION

The lack of effective treatments for neurodegenerative diseases such as Alzheimer's disease (AD) dementia underscores the challenges in identifying the critical pathways for neurodegeneration and novel targets of treatment [1]. Large proteomic analyses of brains and cerebrospinal fluid from patients with AD dementia implicate astrocyte and microglial cell activation in AD pathogenesis [2]. This is not surprising given the active proinflammatory signaling pathways in the brains of patients with not only AD [3,4] but also Parkinson's disease (PD) [5,6], epilepsy [7,8], and cerebral ischemia [9,10]. *Neuroinflammation* is normally a regulated, protective mechanism mediated by astrocytes and microglia in response to brain injury or pathogens [11]. The apolipoprotein E4 allele (*APOE4*) is the strongest genetic risk factor for late-onset AD and there is evidence of chronic unresolved activation of proinflammatory pathways in the *APOE4* dementia brain. The molecular mechanisms leading to neurodegeneration, however, are still unclear. We have recently identified that *APOE4* increases the activation of Ca^{2+} -dependent phospholipase A2 activation (cPLA2) as potential mechanism

for the greater astrocyte activation and neuroinflammation with *APOE4* [12^{***}]. In this review, we discuss the regulation of eicosanoids as mediators of neuroinflammation and novel advances to monitor brain fatty acid metabolism, setting the stage for testing small molecules that can mitigate brain inflammation by inhibiting cPLA2 activation.

EICOSANOIDS: LIPID MEDIATORS OF NEUROINFLAMMATION

In response to extracellular signals of inflammation, cells produce lipid mediators derived from the metabolism of free long-chain polyunsaturated fatty acids (PUFAs). The human brain contains high levels

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KEY POINTS

- *APOE4* associates with greater activation of cPLA2 in the dementia brain and pronounced eicosanoid signaling features consistent with unresolved inflammation.
- Reducing cPLA2 activation in the brain restores BBB integrity and rescues behavioral deficits in animal models.
- AA and DHA PET imaging may provide important tools to study the uptake of these fatty acids decades before the onset of clinical disease and can guide the efficacy of cPLA2 inhibitors.
- Dietary interventions focused on increasing the consumption of DHA and EPA have the best chances of reducing neuroinflammation when started before the onset of neurodegeneration and BBB breakdown.

of PUFAs esterified at the *sn*-2 position of membrane phospholipids, the levels of which are tightly regulated [13]. PUFAs have important roles during development – the infant brain rapidly accumulates arachidonic acid (AA, an ω -6 fatty acid) and docosahexaenoic acid (DHA, an ω -3 fatty acid) which serve as important secondary messengers during neurotransmission in the developing brain, in addition to mediating inflammation [14,15]. Eicosapentaenoic acid (EPA, another ω -3 fatty acid) is present at lower amounts but has strong anti-inflammatory activity [16]. Improper regulation of the balance in PUFAs has been linked to brain diseases [17], but our understanding of the critical mechanisms that underlie brain PUFA dysregulation are still evolving. Among the key enzymes that trigger neuroinflammation, cPLA2 seems to play a key role; the liberation of AA from phospholipids by activated cPLA2 can lead to conversion to proinflammatory mediators (Fig. 1). Prostaglandins, leukotrienes (LTs) and pro-resolving lipids derived from PUFAs have roles in the pathogenesis of neurodegeneration.

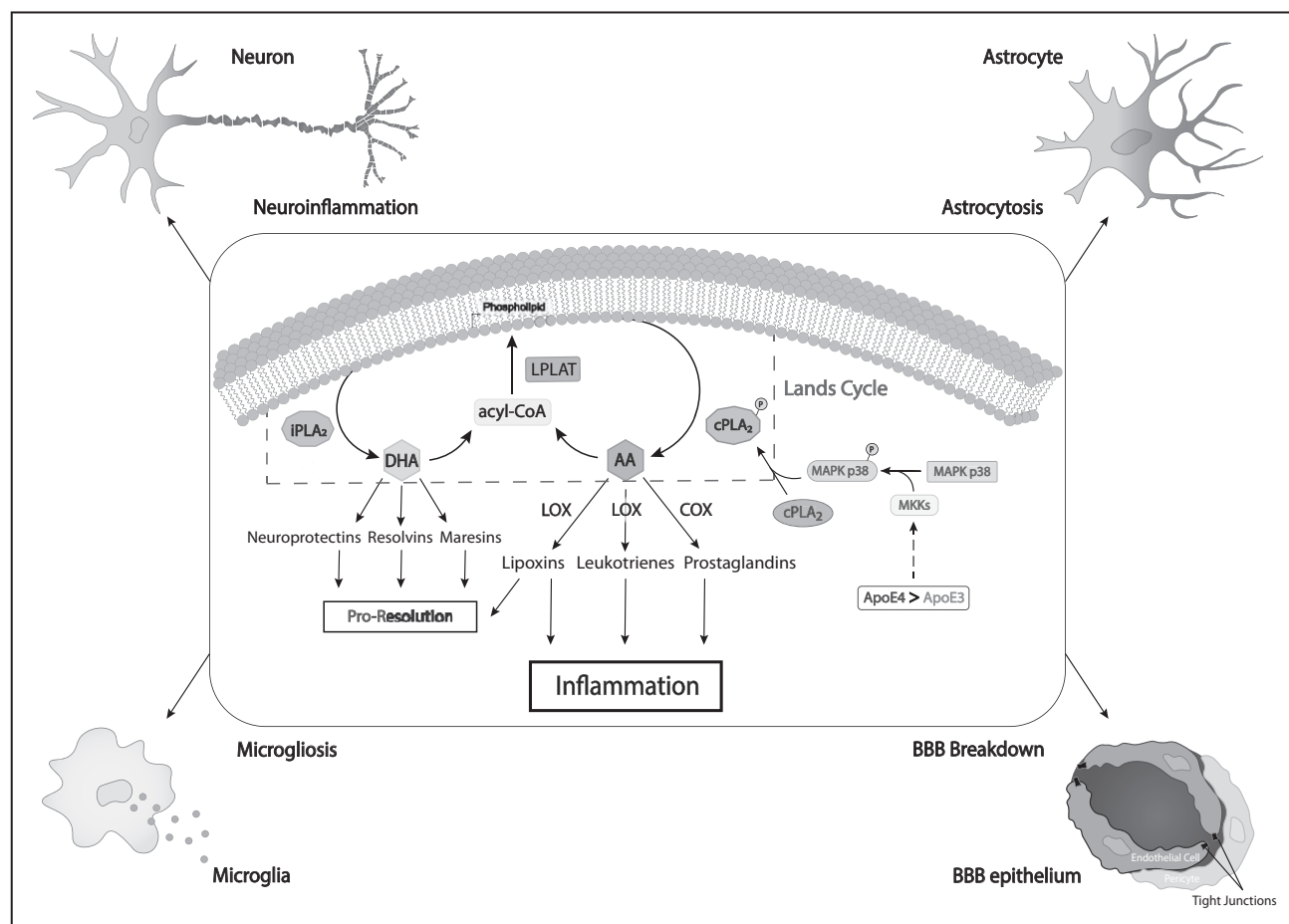


FIGURE 1. An illustration of the Lands cycle and how increased cPLA2 activation can liberate free AA which can be converted to proinflammatory mediators. cPLA2 activation has multiple cell specific effects on the different brain cells and is accentuated by *APOE4*. *APOE4*, apolipoprotein E4; cPLA2, Ca²⁺-dependent phospholipase A2.

Prostaglandins

Prostaglandins, a class of proinflammatory lipid mediators, are derived from the oxidation of AA by cyclooxygenases (COX) and subsequent metabolism by synthases. Although prostaglandins signaling is important as a response to insults, overproduction of these molecules in response to inflammatory signals is implicated in cellular processes involved in the initiation and outcome of AD and PD [18–20]. The molecular processes linking production of prostaglandins to these effects, however, are still not well understood. Recently, prostaglandin E₂ (PGE₂), which is elevated in cancer and neurological damage, was found to have possible effects on the integrity of blood-brain barrier (BBB) vasculature by signaling pericytes into a nonadherent state in vitro [21]. Furthermore, prostaglandins are also believed to be implicated in AD progression and stroke. In transgenic mouse models of AD, up-regulation of PGE₂ synthesis by COX2 was also found to induce the formation of presenilin via a tumor necrosis factor α (TNF α)-dependent pathway, leading to deposition of amyloid beta (A β) and neuronal death [22]. PGE₂ production may also play a role in promoting neuronal degradation in PD. For example, overexpression of microsomal PGE₂ synthase was found to exacerbate PD pathology in a neurotoxin-induced animal model of PD [23]. Prostaglandin D₂ (PGD₂) synthase was found as a potential biomarker for PD [24] which suggests that regulating expression of the enzyme plays a role in the control of PGD₂-mediated inflammation response in PD development, though it is important to note that expression level is not necessarily indicative of activity.

Leukotrienes

LTs represent another distinct class of proinflammatory AA metabolites generated by the body in response to injury, although their biosynthesis is not dependent on COX enzymes. For example, leukotriene A₄ (LTA₄) is generated by 5-lipoxygenase (5-LOX) mediated metabolism of AA, which is rapidly converted to LTB₄ or LTC₄. In vitro, LTA₄ has been shown to exacerbate A β toxicity in neurons; however, the mechanism driving this effect remains unclear as LTA₄ is rapidly metabolized under physiological conditions and its receptors have yet to be defined [25,26]. LTB₄ is known to be more chemically stable than LTA₄ and is regarded as one of the most potent endogenous chemoattractants, promoting inflammation via the recruitment and activation of leukocytes and microglia through BLT1/2 receptors [27]. These LTB₄-induced effects are driven by the induction of mitogen-activated protein kinase and PI3/AKT pathways, which in microglia, result in

greater production of the proinflammatory cytokines involved in AD-related chronic neuroinflammation such as interleukin 1 β (IL-1 β) and TNF- α [28]. Elevation of LTB₄ has also been associated with biomarkers for AD (A β overexpression, tau phosphorylation, and neuroinflammation) in a mouse model [29]. Blockage of the LOX pathway in transgenic mice was shown to mitigate AD progression by modulating neuroinflammation and reducing tau pathologies, which lead to rescue of synaptic integrity and cognitive function [30]. Together, these findings implicate LTs as likely promoters of cognitive decline and dementia by fueling chronic neuroinflammation and potentiating AD pathologies.

Proresolving lipid mediators

Although most AA derivatives are largely proinflammatory, there exists a protective arm of the AA-metabolic pathway mediated by 5-, 12- and 15-LOX enzymes which generate a class of pro-resolving mediators called lipoxins. These bioactive lipids not only exert anti-inflammatory activities by attenuating leukocyte infiltration and the release of reactive oxygen species (ROS) and cytokines, but also promote the resolution pathway by stimulating macrophage uptake of apoptotic neutrophils and debris that contribute to sustained inflammation [31]. These actions are mediated by the binding of lipoxins to formyl peptide receptor 2, a promiscuous G-protein coupled receptor (GPCR) found on the surface of various cell types. In the context of AD, lipoxin A₄ (LXA₄) has been shown to reduce neurotoxin-induced neuronal apoptosis in human neuroblastoma cells [32] and ameliorate AD pathologies. The later has been demonstrated by its ability to preserve cognitive function across multiple disease models by enhancing the phagocytic activity of microglia [33,34].

In addition to lipoxins, three other distinct classes of lipid mediators derived from DHA also possess anti-inflammatory and pro-resolving capabilities: the D-series resolvins, maresins, and neuroprotectins. In the metabolic pathway, DHA is first oxidized by LOXs to form hydroxylated intermediates, which are then converted to resolvins by 5-LOX or maresins and neuroprotectins by soluble epoxide hydrolase (sEH). Much like lipoxins, these ω -3 PUFA metabolites counterbalance the inflammatory response by reducing cytokine release and oxidative stress through GPCR binding across several cell types. Recent in vitro studies demonstrated that resolvin D1 (RvD1) and maresin 1 (MaR1) down-regulate A β -induced inflammation in human microglia and revealed how MaR1 stimulates microglial phagocytosis of A β [32]. Furthermore, clinical

investigations in patients with dementia suggest that a relationship exists between cerebrospinal fluid levels of RvD1 and cognitive function [35]. Neuroprotectin D1 (NPD1) is like DHA-derived resolvins in that it is also a product of sEH metabolism which promotes resolution of inflammation; however, it is distinct from other lipid mediators in that its neuroprotective effects have been well documented. Severely depleted in certain areas of the AD brain [36], NPD1 normally exerts its neuroprotective activity by downregulating A β secretion, upregulating neurotrophic soluble amyloid precursor protein α , and preventing apoptosis of microglia and neurons [37,38].

Postmortem analyses of human brain tissue have shown lower levels of DHA in the AD brain, which consequentially affects the biosynthesis of DHA's proresolving lipid metabolites [36,39]. Similar to DHA, ω -3 fatty acids such as EPA and docosapentaenoic acid (DPA) also serve as precursors to a diverse group of pro-resolving mediators, and EPA has been linked to improved cognitive function in AD [40]. EPA is metabolized by LOXs to generate LXA5, LXB5, and E-series resolvins, and DPA is metabolized in a similar manner to DHA to yield proresolving mediators. Although EPA and DPA metabolites have demonstrated potent proresolving activities, their neuroprotective potential has not been widely studied [41]. Collectively, the proresolving and neuroprotective functions of the lipid mediators described in this section highlight PUFA metabolism as a critical process for maintaining inflammatory homeostasis in the central nervous system, where dysregulation can influence the onset and progression of neurodegenerative diseases such as AD.

LIPID UPTAKE AND RECYCLING

Because PUFAs are essential precursors to lipid mediators, it is important to tightly regulate their levels within membranes and inside the cell to maintain proper function. Although DHA and EPA can be synthesized from other fatty acids (e.g., α -linolenic acid), this process is limited in humans and the exchange/transport of these PUFAs across the BBB is essential to maintaining their proper levels in the brain [42]. In addition to passive diffusion, the trafficking of PUFAs across the BBB is mediated by fatty acid-binding proteins with specificities toward either DHA, EPA, or AA [43,44], and through binding of lysophosphatidyl choline (LPC) DHA to the MFSD2A transporter at the BBB [45]. Once inside the brain compartment, the esterification of these fatty acids via the Lands Cycle (Fig. 1) traps them in the brain and facilitates their accretion. It is important

to maintain the low level of bioactive free PUFAs in the cell and so unesterified fatty acids from the blood, including AA, are introduced into the Lands cycle to maintain basal levels. In this process, PUFAs are readily converted to acyl-CoA and subsequently shuttled to membrane phospholipids through lysophospholipid acyltransferases [46,47], where they are eventually released from membrane phospholipids by receptor-mediated activation of specific phospholipase A (PLA2) enzymes. Exploring factors that affect the fine balance between phospholipid reacylation and hydrolysis reactions in the Lands cycle is important in studying the neuroinflammatory process. Several members of the PLA family release fatty acids from membrane phospholipids in response to cellular signals, including neurotransmission, ATP production, and neuroinflammation [48]. AA, EPA and DHA are found mainly at the *sn*-2 position of brain phospholipids and their release into the cell for further processing to lipid mediators is catalyzed by the Group IV PLA2 enzymes: Ca²⁺-dependent and independent cytosolic PLA2 (cPLA2 and iPLA2, respectively) [49]. DHA and EPA can compete with AA for the *sn*-2 position in phospholipids and an increase in DHA or EPA consumption is associated with a loss of AA, and vice versa [50].

Ca²⁺-dependent phospholipase A2 activation, apolipoprotein E4, and neuroinflammation

Since cPLA2 has greater affinity for the cleavage of AA-containing phospholipids at the *sn*-2 position over DHA or EPA [51], activation of cPLA2 can shift the balance toward more proinflammatory and less resolving mediator production, arguably accelerating the development of neurodegenerative conditions. cPLA2 has been linked to microglial and astrocytic activation during neuroinflammation, playing a role in the induction of LOX activity and ROS production [52]. Although elevated cPLA2 activation has been identified in brains of patients with AD [53], PD [54], epilepsy [55], and cerebral ischemia [56], there is limited research establishing cPLA2 activation states in living brains. In brain autopsy studies, cPLA2 activation colocalizes within astrocytes around A β plaques [57] and in AD mouse models, reducing cPLA2 expression rescues mouse behavioral deficits despite the accumulation of amyloid plaques [58]. There is still a gap in understanding the factors that contribute to the activation of cPLA2 in the living brain when studying neurodegenerative disease.

The APOE4 allele is the strongest genetic risk factor for late-onset AD dementia. However, the link between the altered function of the ApoE4 protein and the pathogenesis of AD is not fully elucidated.

The ApoE4 protein appears to be associated with chronic unresolved neuroinflammation and oxidative stress, which is believed to exacerbate AD pathology [59]. The correlation between *APOE4* and inflammation was clearly demonstrated in the Framingham cohort, where participants with *APOE4* and elevated plasma C-reactive protein levels had a greater risk of developing AD dementia than age and sex-matched *APOE2* and *APOE3* carriers [60]. In hippocampal homogenates from AD brains, *APOE4* carriers had greater levels of thiobarbituric acid-reactive substances as well as lowered catalase activities, glutathione peroxidase, and glutathione levels than tissues from patients carrying the *APOE3* allele [61], markers of elevated oxidative stress.

We hypothesized that *APOE* genotype has disease-stage effects on the metabolism of PUFAs in the brain or their delivery to the brain through the promotion of membrane remodeling via the Lands cycle, affecting DHA transport, or increasing their catabolism. We found increased cPLA2 activation in astrocytes derived from ApoE-targeted replacement mice, whole mouse brain sections, and in human brain tissues [12^{***}]. The mechanisms included greater p38-MAP kinase-mediated activation of cPLA2. We observed greater phosphorylation of cPLA2, which lead to increased production of the AA-derived LT LTB4 (as a downstream effect of AA release from phospholipids), inducible nitric oxide synthase, and ROS. More importantly, inhibition of cPLA2 activity reduced inflammatory signaling and oxidative stress markers in astrocytes. In human brains from *APOE4* carriers, activation of cPLA2 was only seen in those with dementia whereas those without dementia did not. This result suggests that a second insult (such as trauma, systemic inflammation) is required to further enhance the phenotype of increased cPLA2 activation, and perhaps lowering cPLA2 activation in *APOE4* carriers may slow down disease progression.

Ca²⁺-dependent phospholipase A2 activation, brain docosahexaenoic acid delivery and blood-brain barrier breakdown

The BBB is a dynamic metabolic interface with many biological functions that include facilitating the transfer of fatty acids from the blood to the brain parenchyma [62]. The expression of MFSD2A, responsible for the transport of LPC DHA to the brain, is tightly linked to BBB integrity [63] and is responsible for the accretion of DHA in the brain. Mice lacking this transporter have been shown to have attenuated delivery of DHA into the brain and develop microcephaly [45]. Maintenance of the BBB is dependent on the proper regulation of redox

processes within the cells that constitute the cerebral vasculature, and disruption of the balance between ROS generation and elimination leads to BBB leakage [64]. A recent investigation showed that the pathological effect of BBB leakage is apparent even in noncognitively impaired subjects carrying the *APOE4* allele [65^{***}] which may affect DHA brain transport before the onset of clinical disease.

In an in vitro BBB model, AA was found to mediate the permeability of the BBB via its conversion to PGE2 by COX enzymes [66]. In an ex vivo epilepsy model, cPLA2 activity was found to contribute to the upregulation of metalloprotein-mediated BBB dysfunction likely due to the release of AA and its downstream proinflammatory effects, whereas inhibition or genetic knock-out of cPLA2 abolished the vascular leakage associated with BBB degradation [67,68]. These findings relating the *APOE* genotype, cPLA2 activity, and BBB integrity suggest cPLA2 as a viable drug target for mitigating the neuroinflammation and BBB breakdown associated with *APOE4* and AD. Whether preservation of BBB integrity can restore the delivery of DHA to the brain and alter brain PUFA metabolism is an important and unresolved question.

Monitoring polyunsaturated fatty acids uptake in the living brain: role of PET imaging

There is a long preclinical phase in AD lasting decades and is characterized by amyloid accumulation in the brain and subtle cognitive changes [69]. This period is currently viewed as a window for therapeutic interventions before irreversible neurodegenerative changes unfold. New research is required to understand the changes to brain PUFAs during the preclinical and prodromal phases of this disease. Brain PUFA levels are maintained and tightly regulated by increasing or decreasing their uptake from the periphery. Relying on a change in PUFA brain concentrations that appear at the time of clinical disease may lead to missed opportunities to target the disease when the pathology might still be reversible. PUFA brain imaging using positron emission tomography (PET) can provide a tool to monitor brain uptake status during prodromal or preclinical states when PUFA levels are still unchanged. One example is illustrated by computing the incorporation coefficients (k^*) from 1-[¹¹C]-DHA uptake models using PET brain scan images following alcohol withdrawal. PET scan images have demonstrated that brains in the chronic alcohol state extract DHA at much faster rates following systemic injection compared to age-matched controls [70], potentially reflecting a compensatory mechanism used by the

brain to restore its supply of DHA after loss during acute alcohol intoxication.

We have shown that the k^* values for 1- ^{11}C -DHA are significantly higher in a small pool of human subjects possessing the *APOE4* genotype than for noncarriers in their mid-30s. We hypothesized that ApoE4 is affecting the uptake of brain DHA decades before the onset of cognitive decline [71]. Another measure generated by PET imaging is the influx rate (J_{in} , a value related to the total concentration of nonradioactive PUFA in the plasma) and is considered a measure of brain's consumption of fatty acids [46,71]. The reason for increased transport of DHA in younger *APOE4* carriers could relate to greater utilization of DHA as an energy source, or as replacement for DHA lost through increased mitochondrial or peroxisomal oxidation processes [72]. This is not surprising given the greater neural activity observed in younger *APOE4* carriers by resting state functional magnetic resonance imaging studies [73]. With an intact BBB, younger *APOE4* carriers can compensate for increased DHA brain utilization by transporting DHA from the blood into the brain. However, we find evidence for a deficit in brain DHA transport early in the disease process. In the DHA brain delivery pilot trial (NCT02541929), *APOE4* carriers supplemented with DHA prior to the onset of cognitive impairment had smaller increases in CSF DHA levels compared to noncarriers, confirming that *APOE4* may affect DHA transport functions before onset of dementia [74]. These changes are accentuated in *APOE4* carriers with AD dementia [75]. These observations support our efforts in the ongoing

PREVENTE4 trial (NCT03613844) to test whether higher doses of DHA supplementation in *APOE4* carriers without dementia and limited seafood intake can slow down cognitive decline [76].

Another application of PET imaging is directed at measuring AA brain uptake. The k^* values, derived from the modeling of AA brain uptake in relation to the available amount of free AA in the blood, have been used as surrogates to evaluate cPLA2 activity [77,78]. This is based on the vacancy hypothesis where a vacant position at the sn-2 phospholipid hypothesis from increased activation of cPLA2 will drive greater AA brain uptake or k^* . Patients with AD had greater 1- ^{11}C -AA k^* values, by PET scans [78] compared to controls, possibly reflecting the activation of cPLA2. One of the limitations of these studies is the use of ^{11}C radioisotope that has a short half-life ($t_{1/2} = 20$ min) limiting the extent to which they can be tracked in vivo. The use of ^{18}F -functionalized PUFAs, such as 20- ^{18}F fluoro-AA [79,80], presents as a more viable method for translation into the clinic because these tracers have much longer half-lives ($t_{1/2} = 110$ min) than their ^{11}C -functionalized counterparts. We have synthesized ^{18}F fluoro-AA and DHA probes and applied these tracers in preclinical models to study the effect of *APOE* genotype and cPLA2 inhibition on fatty acid uptake into the brain (Unpublished). PET imaging (Fig. 2) allows the estimation of uptake kinetics (k^* and J_{in}), together with regional information, across disease stages and can help select the cognitive domains affected (based on regional uptake changes) and provide an imaging biomarker to guide the efficacy of cPLA2 inhibition in the brain.

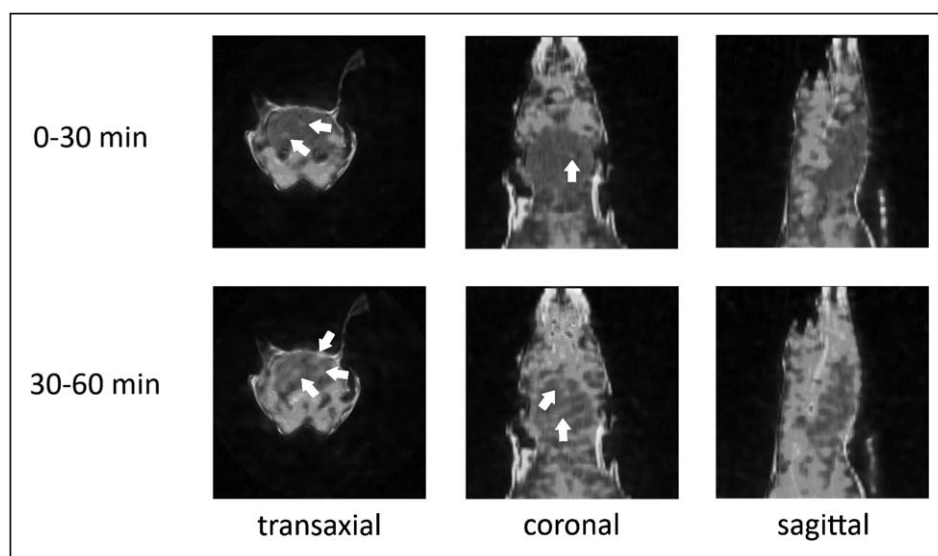


FIGURE 2. PET/MRI image of a mouse 30 min and 60 min after a tail vein bolus injection of 20- ^{18}F fluoro-AA (270.4 μCi dose). Arrows indicate areas where PUFA tracer has accumulated in the brain after uptake over time. PUFAs, polyunsaturated fatty acids.

INTERVENTIONS TO MODULATE NEUROINFLAMMATION

Current therapeutic strategies in the treatment of AD and PD aim to regulate the pathological proteins involved in the onset and progression of these diseases by means of targeting the aberrant proteins or the silencing the genes which express precursors to these proteins [81,82]. Regulating neuroinflammatory responses has also been investigated as a method of preventing disease progression, which may be accomplished by restoring proper metabolism of PUFAs.

Dietary interventions and polyunsaturated fatty acids brain uptake

Supplementation of PUFAs has been considered as a strategy to mitigate the effects of excessive neuroinflammation as deficiencies of PUFAs in the diet have been shown to exacerbate the effects of neurodegenerative pathologies [83]. However, we propose that it is important to determine the effect of disease stage on the delivery of DHA to the brain and the efficacy of DHA supplementation on cognitive outcomes. Before the onset of dementia, we found that carrying the *APOE4* allele associates with a small effect on the delivery of DHA and EPA to the CSF in $\epsilon 4$ patients compared to noncarriers [74], but with no differences in plasma DHA or EPA levels after supplementation. These *APOE4*-associated effects become more pronounced in patients with dementia; we observed lower levels of DHA/AA and EPA/AA in the plasma [75] and CSF [84] of *APOE4* carriers with mild dementia after DHA supplementation compared to noncarriers. The reduced delivery of DHA to CSF in *APOE4* carriers with dementia could explain the lack of improvement in cognitive outcomes after DHA supplementation [85]. These findings highlight the importance of studying the mechanisms of DHA and EPA brain transport not only in different genotype groups but also across different disease states [86].

Inhibition of the eicosanoid pathway

Considering the effects of lipid mediators on neuroinflammation and the growing evidence supporting their involvement in neurodegenerative diseases, enzymes within the PUFA metabolic pathway have gained interest as potential drug targets. For example, the activities of COX and LOX have been probed in the context of AD and PD pathologies using small molecule inhibitors [87–91] because of their roles in facilitating AA metabolism toward the production of proinflammatory signaling lipids. More upstream in the metabolic pathway, cPLA2 is essential for initiating the eicosanoid cascade by cleaving free

AA from membrane phospholipids, and its inhibition has been shown to rescue behavioral deficits in AD mouse models and BBB integrity in epilepsy models [68]. The effects of cPLA2 inhibition on BBB integrity and the brain transport of DHA and AA in *APOE4* carriers are not known and are actively being pursued in our laboratory.

FUTURE APPROACHES TO INVESTIGATING NEUROINFLAMMATION

Solely monitoring the levels of PUFAs and their lipid mediator products in the brain forms an incomplete picture of the neuroinflammatory process because lipid cycling is dynamic and it could take decades for lipid levels to change, meaning additional information is needed to identify this form of metabolic stress before irreversible neurodegeneration occurs. PET imaging can be used to address this gap by allowing the study of fatty acid brain uptake by different disease stage, *APOE* genotype, nutritional status and other factors that affect brain fatty acid uptake. More importantly, this provides a tool to assess the response of prospective treatments. Developing small molecules that can block brain cPLA2 activation provides a considerable opportunity to halt or slow down brain inflammation in neurodegenerative diseases such as AD and PD.

CONCLUSION

Identifying the critical mechanisms by which brain eicosanoid activation is promoted could provide significant insight into AD dementia and enable novel target identification for therapies not only for AD but also related neurodegenerative diseases. Mitigating or preventing neurodegenerative disease may involve control of neuroinflammation in addition to dietary modification. To that end, current progress in pharmacological interventions has shown promise as demonstrated by the development of small molecules targeting key enzymes involved in the uptake of PUFAs and their subsequent conversion to lipid mediators of inflammation. Such interventions provide newer therapeutic strategies for neurodegenerative states.

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Conflicts of interest

There are no conflicts of interest.

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- of special interest
- of outstanding interest

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